

Notation and Lexical Conventions

In the grammar below, $[X]$ means that X is optional, and $\{X\}$ means that X may be repeated zero or more times. Text enclosed in quotation marks represents a terminal symbol.

Spaces and tabs are discarded by the tokenizer. A line terminator following a line containing tokens produces a **NEWLINE** token. Blank and whitespace-only lines are discarded and do not produce **NEWLINE** tokens. The tokenizer produces **EOF** after the final input character; it does not insert an implicit **NEWLINE** before **EOF**.

Statements

<code><program></code>	$\rightarrow$ <code><statements> EOF</code>
<code><statements></code>	$\rightarrow$ <code><statement> NEWLINE {<statement> NEWLINE }</code>
<code><statement></code>	$\rightarrow$ <code><for-statement> <assignment-statement> <print-statement></code>
<code><for-statement></code>	$\rightarrow$ <code>"for" (</code> <code><assignment-statement> ; <rel-expr> ;</code> <code><assignment-statement>)</code> <code>{ NEWLINE <statements> }</code>
<code><assignment-statement></code>	$\rightarrow$ <code><id> = <rel-expr></code>
<code><print-statement></code>	$\rightarrow$ <code>"print" <rel-expr></code>

Expressions

<code><rel-expr></code>	$\rightarrow$ <code><rel-term> [<equality-op> <rel-term>]</code>
<code><rel-term></code>	$\rightarrow$ <code><rel-primary> [<ordering-op> <rel-primary>]</code>
<code><rel-primary></code>	$\rightarrow$ <code><arith-expr></code>
<code><arith-expr></code>	$\rightarrow$ <code><arith-term> {<add-op> <arith-term> }</code>
<code><arith-term></code>	$\rightarrow$ <code><arith-primary> {<mult-op> <arith-primary> }</code>
<code><arith-primary></code>	$\rightarrow$ <code>[<sign>] <arith-atom></code>
<code><arith-atom></code>	$\rightarrow$ <code><id> <integer> (<rel-expr>)</code>
<code><id></code>	$\rightarrow$ <code>[_a-zA-Z][_a-zA-Z0-9]*</code>
<code><integer></code>	$\rightarrow$ <code>[0-9]⁺</code>
<code><sign></code>	$\rightarrow$ <code> + -</code>
<code><equality-op></code>	$\rightarrow$ <code> "==" "!="</code>
<code><ordering-op></code>	$\rightarrow$ <code> ">" ">=" "<" "<="</code>
<code><add-op></code>	$\rightarrow$ <code> + -</code>
<code><mult-op></code>	$\rightarrow$ <code> * / %</code>